METRIC

MIL-PRF-32138
20 November 2003

PERFORMANCE SPECIFICATION

BOOTS, CONNECTOR CORROSION PROTECTION, GEL-ENCLOSED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for re-enterable, gel-enclosed, connector corrosion protection boots used to provide environmental protection for radio-frequency waveguides used in shipboard and exposed environment applications.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests.

(Copies of this document are available online at http://astimage.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data, that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ELECTROCHEMICAL COMMISSION (IEC)

68-2-11 - Environmental Testing - Part 2: Tests. Test Ka: Salt Mist.

(Copies of this document are available online at www.iec.ch or from IEC Central Office, 3 rue de Varembé, P.O. Box 131, CH – 1211 Geneva 20, Switzerland.)

SAE INTERNATIONAL

J1777 - General Environmental Considerations for Marine Vehicles.

J1976 - Outdoor Weathering of Exterior Materials.

(Copies of these documents are available online at www.sae.org or from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials and components. The boots shall have a preformed polyethylene construction that, when secured, encloses the waveguide connection. The inner portion of the boot shall have a silicone gel, to surround the connector and create an environmental barrier.

3.2.1 Basic material. The boots shall be fabricated from flame-retardant, ultraviolet (UV) crosslinked polyolefin with a medium viscosity, crosslinked silicone gel inclusion. The inclusion acts as an environmental barrier between the boot and the waveguide surface.

3.2.2 Tie straps. Three black plastic tie straps shall be included to secure the boot assembly about the waveguide connection. The tie strap shall employ a fastener method, which permits opening and reuse of the tie strap.

3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Dimensions. The corrosion protection boots shall be fabricated so as to provide adequate clearances for assembly and removal. The following dimensions, as identified in Figure 1, shall be provided when ordering corrosion protection boots (see 6.2):

Location	Measurement (mm)
A	Waveguide Height
B	Waveguide Width
C	Flange Height
D	Flange Width
E	Flange Thickness

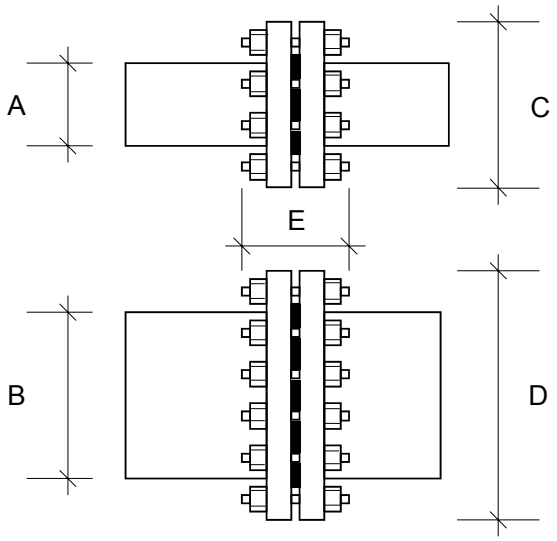


FIGURE 1. Flange dimensions.

3.4 Performance characteristics.

3.4.1 Temperature. The boots shall remain fully functional for the temperature range of -25 to 65 °C.

3.4.2 Solar radiation. The boots shall remain fully functional when exposed to solar radiation.

3.4.3 Salt spray. The boots shall provide protection from and be unaffected by salt spray.

3.4.4 Greenwater loading. The boots shall provide protection from and be unaffected by greenwater loading of 42 kPa (kilopascals).

3.4.5 Stack gases. The boots shall provide protection from and be unaffected by stack (engine exhaust) exhaust gases.

3.4.6 Service life. The design service life for the specified operating environment shall be 8 years.

3.4.7 Shelf life. The shelf life of kits shall be not less than 60 months when stored in a sheltered space having a controlled environment.

3.5 Assembly. The boots shall be provided as kits, which shall incorporate all materials required for one complete installation. Each kit shall include the mating boot components and three tie straps. Kits shall be individually packaged.

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3.6 Workmanship. Boots shall be free of all foreign matter. Boots shall be uniform in quality and free from irregularities or defects that could adversely affect performance, durability, or service life.

3.7 Labeling. The kits shall be labeled according to operating frequency bands of the waveguide covered by the kit (i.e., x-band).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article inspection shall be performed on one complete boot kit when a first article sample is required (see 3.1). This inspection shall include the examination specified in 4.4 and the tests of 4.5 (see Table I).

4.3 Conformance inspection. The conformance inspection shall consist of the examination specified in 4.4 and the tests of 4.5.1.2, 4.5.1.3, and 4.5.3 (see Table I). Conformance inspections shall be as specified in the contract or request for proposals.

4.4 Examination. Each kit shall be visually inspected for compliance with the requirements of 3.2, 3.2.1, 3.2.2, 3.3, 3.5, and 3.6 (see Table I). At the contractor's option, portions of the visual inspection may be performed by automated optical and electronic sensors. When automated inspections are utilized to support first article and conformance inspections, sufficient instrumentation shall be provided so as to document the performance of the automated sensors. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE I. Requirements and verification methods.

Requirement Test	Requirement Paragraph	1st Article Test Paragraph	Conformance Inspection Paragraph
Basic material	3.2.1	4.5.1.1	N/A
Tie straps	3.2.2	4.5.1.2	4.5.1.2
Assembly	3.5	4.5.1.3	4.5.1.3
Dimensions	3.3	4.5.2	N/A
Temperature	3.4.1	4.5.3	4.5.3
Solar radiation	3.4.2	4.5.4	N/A
Salt spray	3.4.3	4.5.5	N/A
Greenwater loading	3.4.4	4.5.6	N/A
Stack gases	3.4.5	4.5.7	N/A

4.5 Methods of inspection.

4.5.1 Visual inspection.

4.5.1.1 Basic material. The cross-linked polyolefin material shall be visually inspected to meet the requirements of 3.2.1 (see 4.4).

4.5.1.2 Tie straps. The tie straps shall be visually inspected and tested for the ability to be secured and then reopened (see 3.2.2 and 4.4).

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4.5.1.3 Assembly. Kits shall be visually inspected to ensure that all specified components are present. Where automated processes are employed for the assembly of kits, electro-optical devices may be used to perform the inspection (see 3.5).

4.5.2 Dimensions. The dimensions of the boots shall be measured and compared to the dimensions specified in 3.3.

4.5.3 Temperature. Temperature tests shall be in accordance with MIL-STD-810, methods 501 and 502 (see 3.4.1).

4.5.4 Solar radiation. Solar radiation tests shall be in accordance with MIL-STD-810, method 505, procedure II (see 3.4.2).

4.5.5 Salt spray. Salt spray tests shall be in accordance with MIL-STD-810, method 509, or IEC 68-2-11 with a duration of 48 hours (see 3.4.3).

4.5.6 Greenwater loading. Greenwater loading shall be tested by applying a hydrostatic pressure equal to the design load for 15 minutes. A visual inspection shall reveal no water intrusion under the sleeve (see 3.4.4).

4.5.7 Stack gases. The completed assembly shall be tested and shown to be unaffected by the exhaust produced by the combustion of distillate fuels in gas turbines and internal combustion engines. Test procedures shall be in accordance with SAE J1777 and J1976 (see 3.4.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boots are intended for use in exposed, unsheltered environments aboard surface ships, and shore installations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Waveguide dimensions and frequency band (if not previously supplied) (see 3.3).

c. Quantity required.

d. When first article is required (see 3.1).

e. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

Greenwater

Radio-frequency waveguides

Salt spray

Shipboard

Custodians:
Navy - SH
Air Force - 99

Preparing Activity:
Navy - SH
(Project 2090-0119-000)

Review Activity:
Air Force - 71
DLA - CC

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